

Scenario	Primary Metric	Key Finding	RQ	Deployment Implication
Key Exchange	Median latency (ms)	ECDH: 0.28 ms · ML-KEM: 0.49 ms · RSA: 52 ms	RQ1	ML-KEM viable ECDH replacement; RSA is unsuitable for high-frequency session establishment in this environment
Hash Functions	Throughput (MB/s) at 1 MB	SHA-256: 446.92 MB/s · SHA3-256: 361.81 MB/s · SHAKE256: 362.16 MB/s (~19% gap)	RQ3	Sponge functions practical for PQC pipelines; gap immaterial for short messages
AES Encryption	Throughput (MB/s) at 100 MB	AES-128: 1,234 MB/s · AES-192: 1,173 MB/s · AES- 256: 1,176 MB/s (~5% gap)	RQ3	AES-256 preferred for max security; marginal throughput penalty at scale
Hybrid Workflow	Total median latency (ms)	0.4708 ms total; Kyber512: 69.2%, ECDH: 29.1%, KDF: 1.0%; AES unaffected	RQ2	Hybrid ECDH + ML-KEM operationally practical sub- millisecond; adds forward secrecy + PQC resistance

Table V.6: Cross-Scenario Findings Summary